



SILUNI WIJESINGHE

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PROFILE

Final-year IT undergraduate at the University of Moratuwa (CGPA: 3.78/4.00), with 6 months of hands-on industry experience building large-scale, multi-client ERP systems at Agrithmics. Delivered production-ready features across procurement, payroll, inventory, and manufacturing modules for international projects. Currently conducting final-year research on automated Ayurvedic medicinal leaf analysis using machine vision and deep learning. Proficient across the full stack with experience in SQL optimization, and Agile delivery, with a strong ability to adapt to evolving requirements and consistently produce clean, maintainable code.

EXPERIENCE

Intern Full Stack Software Engineer
Agrithmics (Pvt) Ltd, Sri Lanka

Feb 2025 - Aug 2025

- **Enterprise ERP - Full-Stack Feature Delivery:** Built full-stack features across 4 international ERP platforms - Ispahani, Lankem, Hayleys, and HNB Farmer System using C#/.NET and React, spanning procurement, payroll, inventory, and manufacturing modules.
- **Feature Engineering:** Delivered 10+ screens end-to-end including Quotation Management screen with item-wise multi-vendor selection logic and automated comparison scoring, reducing manual procurement decision time by consolidating vendor evaluation onto a single screen; implemented Excel export for bulk quote downloads.
- **Background Process Debugging:** Traced and fixed a critical execution bug in the Sick Leave Scheduler causing incorrect payroll processing in production; resolved by analysing scheduling logic and service execution flow within a single sprint.
- **Reporting & SQL Optimisation:** delivered 10+ dynamic reporting modules including a Maternity Leave Weekly Payment Report with automated PDF generation, configurable date-range validation, and per-column dynamic generation, reducing manual HR report preparation time by eliminating a recurring SQL data-grouping error that had affected production for two payroll cycles.
- **Codebase Quality & Performance:** Refactored 5+ legacy query modules following senior code review; replaced N+1 query patterns with optimised SQL joins and indexed stored procedures, measurably reducing report load times on data sets exceeding 10,000 records.
- **Bug Fixing & Code Quality:** Resolved production bugs across inventory, payroll, attendance, and manufacturing screens; refactored codebase following supervisor feedback to improve query efficiency and reduce redundancy.
- **DevOps & Agile Delivery:** Managed sprint tasks via Azure DevOps - Git branching, pull requests with peer review, unit tests, code reviews

PROJECTS

CentralSync - Cross-Platform Inventory Management System [\[Frontend\]](#) [\[Backend\]](#) [\[Mobile\]](#)

Rootcode Labs | React.js, React Native, Spring Boot, MySQL

- Architected and delivered end-to-end inventory modules (item, stock, reporting) across web and mobile platforms.
- Built REST APIs with JWT authentication and RBAC, supporting secure multi-client operations.
- Implemented real-time low-stock alerting system with configurable thresholds and automated PDF/Excel reporting.
- Designed relational schema with optimistic locking to handle concurrent stock updates across distributed clients.

Pathwise - AI-Powered Career Guidance Platform [\[Frontend\]](#) [\[Backend\]](#)

Idealize 2025 | Next.js, Spring Boot, MySQL, Generative AI

- Developed coach-session booking system with conflict detection and scheduling logic.
- Built real-time WebSocket-based messaging module with persistence and read receipts.
- Integrated full-stack flow from Next.js frontend to Spring Boot APIs and MySQL database.

Machine Vision for Ayurvedic Species Identification & Leaf Health Assessment [\[GitHub\]](#)

Final Year Research (Ongoing) | Python, OpenCV, TensorFlow, scikit-learn

- Leading the Compound Leaf Analysis module, designing a machine vision pipeline to classify morphologically similar Ayurvedic species and assess plant health.
- Developing a comparative framework between handcrafted morphological feature extraction (shape, texture) and deep learning models (CNNs).
- Addressing real-world challenges such as visual similarity and data scarcity in Sri Lanka’s herbal medicine supply chain.

LankaGrid-Forecaster - Short-Term Energy Load Prediction [\[GitHub\]](#)

Individual Project | Python, XGBoost, SHAP, Streamlit, scikit-learn, Pandas

- Built time-series forecasting model using XGBoost to predict national electricity demand at 15-minute intervals.
- Achieved $R^2 = 0.9991$ with 5-fold cross-validation; identified key drivers using SHAP explainability.
- Deployed interactive Streamlit dashboard for real-time scenario simulation

Human Rescue Rover - Embedded Systems

C++, Microcontrollers, Wireless Communication, Sensors

- Built an autonomous rescue rover using microcontrollers with ultrasonic sensors for real-time obstacle avoidance.
- Implemented sensor fusion logic to improve navigation accuracy in dynamic environments.
- Developed wireless video streaming for remote monitoring in simulated rescue scenarios.
- Demonstrated reliable operation in a constrained, real-time environment.

TECHNICAL SKILLS

- Languages :** Java, C#, JavaScript, TypeScript, Python, C++
Frontend : React, Next.js, React Native, Tailwind CSS, HTML/CSS
Backend : Spring Boot, .NET Framework, RESTful APIs, Entity Framework
Datbase : MySQL, Microsoft SQL Server
ML & Data : XGBoost, TensorFlow, scikit-learn, OpenCV, SHAP, Streamlit, Pandas
DevOps & Tools : Azure DevOps, Git, GitHub, Postman, Agile/Scrum

EDUCATION

BSc. (Hons) in Information Technology	2022 – Present
University of Moratuwa CGPA: 3.78/4.00	
Dean’s List: L1S2 (SGPA 3.92) · L2S1 (SGPA 4.00) · L2S2 (SGPA 3.96)	
G.C.E. Advanced Level — Physical Science	2020
Anula Vidyalaya, Nugegoda Z-score: 1.7467 2As & 1B	
G.C.E. Ordinary Level	2017
Ananda Balika Vidyalaya, Kotte 9 A’s	

ACHIEVEMENTS

- Semi-finalist - Idealize 2025
- Semi-finalist -InsipiHer
- Participant - Coderush, Devthon, MoraXtreme:

REFERENCES

Mr. Evantha Dissanayake Senior Software Engineer, Agrithmics (Pvt) Ltd evantha@agrithmics.lk	Mr. Vidath Dahanayake Software Engineer, Oklo (Pvt) Ltd vidathd@okloworld.com
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